

Delivery Index

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Virtual Battery Factory · case t7_r1 · 2026-08-25

Delivery Index — Virtual Battery Factory Case t7_r1

Document: VBF-T7R1-IDX-01 · Generated 2026-08-25 · Signature: Prepared _____ / Reviewed _____ / Approved _____

- Case name: t7_r1 — HEV battery design (energy density ≥ 327.18 Wh/kg, 4C fast charge without lithium plating, SEI ≤ 550 nm after 100 cycles @ 45 °C, nail penetration 10 W without thermal runaway)
- Numbering scheme: `VBF---`; case ID `t7\_r1` → alphanumeric token T7R1 (non-alphanumeric characters removed per numbering rule); SEQ-NO two digits from 01
- Generation date: 2026-08-25
- Result: achieved (final entry verdict; V11-final passes 4/4 criteria at DFN precision)

Document Code Reference

Document code	Meaning	File
DS	Specification	design_spec.md
BOM	Bill of Materials	bom.xlsx
DSH	Datasheet	datasheet.md
CALC	Calculation sheet	calc.xlsx
DVPR	Design verification report	dvpr.md
DFMEA	Failure analysis	dfmea.md
IDX	Delivery index	delivery_index.md

File List

File name	Number	Format	Source description
design_spec.md	VBF-T7R1-DS-01	md	generated per deliverable-design-spec spec; all values from parameter-set dump + R5 tool outputs
design_spec.pdf	VBF-T7R1-DS-01	pdf	md → PDF release via reportlab one-off script
bom.xlsx	VBF-T7R1-BOM-01	xlsx	openpyxl; component masses from calc-energy layer masses + literature split fractions (annotated)
bom.pdf	VBF-T7R1-BOM-01	pdf	xlsx → PDF release via reportlab
datasheet.md	VBF-T7R1-DSH-01	md	generated per deliverable-datasheet spec
datasheet.pdf	VBF-T7R1-DSH-01	pdf	md → PDF via reportlab
calc.xlsx	VBF-T7R1-CALC-01	xlsx	openpyxl; five sheets: inputs / capacity-energy / energy density / N-P-mass / process parameters; formula + source columns
calc.pdf	VBF-T7R1-CALC-01	pdf	xlsx → PDF via reportlab
dvpr.md	VBF-T7R1-DVPR-01	md	generated per deliverable-dvpr spec (virtual test version)
dvpr.pdf	VBF-T7R1-DVPR-01	pdf	md → PDF via reportlab
dfmea.md	VBF-T7R1-DFMEA-01	md	generated per deliverable-dfmea spec (qualitative)
dfmea.pdf	VBF-T7R1-DFMEA-01	pdf	md → PDF via reportlab
delivery_index.md	VBF-T7R1-IDX-01	md	this file, per deliverable-package spec
delivery_index.pdf	VBF-T7R1-IDX-01	pdf	md → PDF via reportlab (blueprint-style cover)
report.html	VBF-T7R1-DS-02	html	`bda render` output (seven-section HTML report from log.jsonl; ancillary of DS)

Notes

- CAD structure model (cell_model.stl): not produced — optional deliverable, requires user clarification for form/presentation; none given in this headless run.
- PDFs generated with a one-off reportlab script (`.venv`); cover style follows engineering-blueprint colors.

- All conclusion-grade values in this package trace to tool output files under `cell/` (run-pyamm / calc-energy / run-tr) or to the Chen2020 parameter-set dump; no value written from memory.
- True first-principles endorsement skipped: `real_compute = false` (entry 0 meta; see endorse entry in log.jsonl).